

SAFETY QUIZ: CHE382-Chemical Engineering Laboratory II

Please answer the following questions testing your knowledge from 6 selected chapters of the JT Baker Hazardous Chemical Safety document; Toxic Chemicals, Flammables, Corrosive Chemicals, Reactive Chemicals, Insidious Hazards and Compressed Gases. Be brief but concise in your answers. Open book/open notes and team discussion is allowed but answers must be in your own words. (5 points per question)

1. What is the general definition of a toxic chemical?

A: The U.S. Environmental Protection Agency or EPA defines a toxic chemical as any substance which may be harmful to the environment or hazardous to your health if inhaled, ingested or absorbed through the skin.

2. Name the general types of damage that a toxic chemical can cause:



A: chemicals can cause many types of toxicity by a variety of mechanisms

3. Name two parameters that can affect toxicity:



A: stress and amount of chemical is toxic

4. What is the DOT (Department of Transportation) definition of a nonflammable compressed gas?

A: compressed gases are defined by DOT as those nonflammable materials which have a psia greater 40 at 70 °F or a psia greater than 104 °F at 130 °F, or those flammable materials which have a psia than 40 at 100 °F

5. a) How does one determine how much material is remaining in a non-liquefied gas cylinder?

A: in a non- liquefied gas cylinder the amount of material remaining is directly proportional to the pressure inside the cylinder which is the cylinder pressure gauge reading.

b) On page 13-10, which pressure gauge on the two-stage regulator will give you the cylinder pressure reading?

A: the reading is on the 500 – 3000 one right next to the compressed gas tank.

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6. Other than a single compressed gas, what else can be stored inside a compressed gas cylinder?

A: Gas mixture, liquified gas, non-liquified gas, and gas dissolved in liquid or solid can be stored inside a compressed gas cylinder.

7. Why do insidious hazards represent a problem to workers?

A: because insidious hazards can cause local or systemic, acute or chronic effects and worker may never be aware that a problem exists until chronic and systemic poisoning has occurred.

8. What are the four general categories of insidious hazards?

A:

1. Unanticipated chemical reaction due to: peroxide formation, gas formation and etc.
2. unanticipated toxicity
3. improper facility design
4. improper handling of equipment and chemicals

9. Name 2 control measures that can be implemented to reduce insidious hazards in the laboratory.

A:

1. thoroughly train all operators in the hazards of substances commonly used in the laboratory.
2. do not allow stock buildup of toxic, flammable, or corrosive materials.

10. What is the definition of a reactive chemical?

A: reactive chemicals are substance which enter into violent reaction during which the spontaneous liberation of heat and/ or gases is too rapid to be safely dissipated by the surrounding.

11. Name the two groups and their sub-headings used to categorize reactive chemicals:

A:

1. Unstable compounds:
 - explosive and other
2. unstable mixture:
 - explosive such as black power
 - Water reactive compounds
 - Air-sensitive compounds
 - Other mixtures of oxidizing agents with reducing agents

12. List two precautionary measures for handling and/or storing reactive chemicals.

A: isolate reactive chemicals. Store compounds of a class together. Away from other classes.

For water-sensitive chemicals, do not allow water in the storage area, or water type fire extinguishers and sprinklers.

13. How are corrosive chemicals generally defined?

A: corrosive chemicals are those substance that, by direct chemical action, are injurious to body tissue or corrosive to metal. Corrosive injury may be to a minor degree or actual disruption of body tissues.

14. How do corrosive chemicals act on body tissues?

A:

1. Direct chemical reaction
2. dissolution of essential components
3. destruction of protein
4. disruption of cell membranes

15. True or False: Concentrated alkaline solutions have a more corrosive effect on tissue than most strong inorganic acids.

A: true

16. Name the six categories of common corrosive liquids.

A:

1. Inorganic acids
2. organic acids
3. Other inorganics
4. caustic solutions
5. organic solvents
6. other organics

17. What category of corrosive solids presents the greatest hazard because of their frequent use in industry?

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A: corrosive solids has the effects of solid corrosive are largely dependent on their solubility in skin and respiratory moisture.



18. Name the three components that make up the FIRE TRIANGLE:

A: fuel energy, fuel oxidizer, oxidizer energy

19. What category of flammables is the most common source of fires in industry?

A: flammable solvents are the most common source of fires in the industry.

20. List two common safety problems and two safety control measures for flammable solvents.

A:

safety problems

1. a lack of an inadequate grounding to prevent static sparks, generated by the flowing liquids.
2. smoking in the solvent areas, ignition sources present.

Common control measures

1. eliminate courses of ignition,
2. work with and transport solvents only in a well-ventilated area.

BONUS QUESTION:

Where is the eyeglass cleaning station located in the Unit Operations Laboratory (Room 120)?

A: the sink on the west side of the unit operation lab room which has a huge class window to spy on what happen in classroom 124's

Comment Summary

Page 1

1. -5

Local toxicity, systemic toxicity, acute toxicity, chronic toxicity

2. -5

Mode of entry, physical condition, dose, duration, sensitivity, combined effects, stress, others (must list)

Page 4

3. -5

Caustic alkalies